BARATH R

Software Developer | Java · Spring Boot · React

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PROFESSIONAL SUMMARY

Software Developer with 2.8+ years of experience building full-stack applications using Java, Spring Boot, React, SQL, and WebSockets. Currently working at AU-KBC Research Centre, Anna University, where I develop and maintain software platforms supporting 100+ researchers, including real-time applications, role-based access, corpus management, and model development workflows. Experienced in taking applications from development to deployment, with additional hands-on experience in Python, AI/NLP, and Android development.

PROFESSIONAL EXPERIENCE

Software Developer

AU-KBC Research Centre, Anna University - Chennai

February 2024 – Present

Discourse Platform - ***React · Spring Boot · WebSocket · SQL***

Full-Stack Development | Platform used by 100+ Researchers

- Developed and architected a full-stack research platform using React, Spring Boot, WebSockets, and SQL, supporting real-time annotation and model-development workflows for 100+ researchers.
- Built an annotation module supporting manual POS, morphological, chunking, and NER tagging, along with automatic annotation using CRF models; implemented corpus management and file-handling workflows for structured, multi-format datasets.
- Implemented multi-tenant architecture with role-based access for individual users and organizations, using SQL-based data management and secure on-premise storage for application data, datasets, and models.
- Built a model development module enabling researchers to develop, train, and run inference on deep learning models within the platform, integrating annotated datasets directly into model-training workflows.

LCAuthoring Interface - ***HTML · CSS · JavaScript · Java · MongoDB*** | Co-developed

- Developed a multilingual web-based annotation tool using HTML, CSS, JavaScript, and Java, enabling researchers to annotate plain-text and CSV datasets across multiple Indian regional languages.
- Built the full-stack application with an HTML/CSS/JavaScript frontend, Java backend, and MongoDB for storing and managing annotation data.

Anna University Chatbot - ***CRF Models · Rule-based NLP · Java*** | Co-developed

- Co-developed an NLP-powered chatbot using Java and CRF models to answer student queries related to admissions, fee structure, and academic information.
- Integrated Anaphora Resolution to interpret contextual references in multi-turn conversations and improve chatbot response accuracy.
- Combined CRF model predictions with rule-based logic to implement a hybrid approach for intent classification and response generation.

Bhashayatra - AI Translation & Transcription App | ***Java · Native Android*** *· In-house NLP Models* | Live on Google Play Store

- Co-developed and deployed a production Android application supporting real-time translation and transcription across multiple Indian and regional languages.
- Integrated in-house AU-KBC translation and transcription models into the Android application and deployed the product on the Google Play Store.

Research Data Pipeline - ***Python*** *· Web Scraping*

- Developed Python-based web scraping pipelines to extract and process data from diverse online sources for research data collection and NLP corpus development.

PROJECTS

VisionX - Medical X-Ray Display & Enhancement App - *Java · Android TV · OpenCV · Image Processing*
Client Project

- Sole developer of a native Android TV application for a medical client, enabling real-time X-ray image capture, enhancement, and display directly from X-ray machine hardware via cable input.
- Implemented OpenCV-based image processing and custom mathematical techniques to remove salt-and-pepper noise and enhance X-ray image clarity for medical visualization.
- Developed the complete processing pipeline from raw X-ray signal capture and image processing to enhanced real-time display, independently delivering the application for a medical client.

TECHNICAL SKILLS

Languages: Java, JavaScript (ES6+), Python

Frontend: React.js, Tailwind CSS, Three.js

Backend: Spring Boot, REST APIs, WebSockets

Mobile: Android Development (Java), Android TV, Android SDK

AI / ML & NLP: Machine Learning, NLP, CRF Models, Model Training & Fine-tuning, Chatbot Development, Data Annotation, Web Scraping (Python), OpenCV, Image Processing

Databases: SQL, MongoDB

Tools & DevOps: Git, GitHub, Docker, Firebase

EDUCATION

Bachelor of Engineering, Computer Science and Engineering

Jeppiaar Engineering College, Chennai - Anna University | CGPA: 8.3 / 10 | July 2023

Relevant Coursework: Data Structures & Algorithms, DBMS, Web Technologies, Operating Systems, OOP, Software Engineering, Computer Networks

HONORS & AWARDS

- **Naan Mudhalvan Achiever** - Awarded by the Chief Minister of Tamil Nadu for outstanding technical achievement.
- **1st Place, INSTAMIND Hackathon** - Solved a real-world logistics problem for DUNZO under competitive conditions.
- **3rd Place, National Hackathon - JNN Institute** - Built a Waste Management Segregation system at national level.
- **IBM SWMS IoT Project Recognition** - Recognised by IBM for contributions to a Smart Waste Management System IoT project.